

Problem Set

1. Each of the vertices of a regular nonagon has been colored either red or blue. Prove that there exist two congruent monochromatic triangles; that is, triangles whose vertices are all the same color.
2. Let n be a positive integer. In how many ways can one write a sum of (at least two) positive integers that add up to n ? Consider the same set of integers written in a different order as being different. (For example, there are 3 ways to express 3 as $3 = 1 + 1 + 1 = 2 + 1 = 1 + 2$.)
3. Ten points are selected on the positive x -axis, X^+ , and five points are selected on the positive y -axis, Y^+ . The fifty segments connecting the ten points on x^+ to the five points on Y^+ are drawn. What is the maximum possible number of points of intersection of these fifty segments in the interior of the first quadrant?
4. Let n be an integer with $n \geq 2$, and define the sequence $S = (1, 2, \dots, n)$. A subsequence of S is called arithmetic if it has at least two terms and it is an arithmetic progression. An arithmetic subsequence is called maximal if this progression cannot be lengthened by the inclusion of another element of S . Determine the number of maximal arithmetic subsequences.
5. Suppose that two PEA administrators are the only persons who have dial-in access to the Academy's Internet fileserver, which can handle only one call at a time. Each has a 15-minute project to do and hopes to use the fileserver between 4 PM and 6 PM today. Neither will call after 5:45 PM, and neither will call more than once. At least one of them will succeed. What is the probability that they both complete their tasks?
6. Let m and n be positive integers.

There are $\binom{n-1}{m-1}$ ordered m -tuples $(x_1, x_2, \dots, x_m)$ of positive integers satisfying the equation $x_1 + x_2 + \dots + x_m = n$.

There are $\binom{n+m-1}{m-1}$ ordered m -tuples $(x_1, x_2, \dots, x_m)$ of nonnegative integers satisfying the equation $x_1 + x_2 + \dots + x_m = n$.
7. Five regular dice are rolled. What is the probability that the sum of the five numbers shown is equal to 14?
8. Given eight distinguishable rings, find the number of possible five-ring arrangements on the four fingers (not the thumb) of one hand. (The order of the rings on each finger is significant, but it is not required that each finger have a ring.)
9. A triangular grid is obtained by tiling an equilateral triangle of side length n by n^2 equilateral triangles of side length 1 (Figure). Determine the number of parallelograms bounded by line segments of the grid.

10. A fair die is rolled four times. What is the probability that each of the final three rolls is at least as large as the roll preceding it?
11. Tram tickets have six-digit numbers (from 000000 to 999999). A ticket is called lucky if the sum of its first three digits is equal to that of its last three digits. A ticket is called medium if the sum of all of its digits is 27. Let A and B denote the numbers of lucky tickets and medium tickets, respectively. Find $A - B$.
12. Let n be a positive integer. Points $A_1, A_2, \dots, A_n$ lie on a circle. For $1 \leq i < j \leq n$ we construct $\overline{A_i A_j}$. Let S denote the set of all such segments. Determine the maximum number of intersection points can be produced by the elements in S .
13. Let $A = \{a_1, a_2, \dots, a_{100}\}$ and $B = \{1, 2, \dots, 50\}$. Determine the number of surjective functions f from A to B such that $f(a_1) \leq f(a_2) \leq \dots \leq f(a_{100})$. What if f does not need to be surjective?
14. For $\{1, 2, \dots, n\}$ and each of its nonempty subsets a unique alternating sum is defined as follows: Arrange the numbers in the subset in decreasing order and then, beginning with the largest, alternately add and subtract successive numbers. (For example, the alternating sum for $\{1, 2, 4, 6, 9\}$ is $9 - 6 + 4 - 2 + 1 = 6$ and for $\{5\}$, it is simply 5.) Find the sum of all such alternating sums for $n = 7$.
15. Let $\prod_{n=1}^{1996} (1 + nx^{3^n}) = 1 + a_1x^{k_1} + a_2x^{k_2} + \dots + a_mx^{k_m}$,
where $a_1, a_2, \dots, a_m$ are nonzero and $k_1 < k_2 < \dots < k_m$. Find a_{1234} .
16. A triangular grid is obtained by tiling an equilateral triangle of side length n with n^2 equilateral triangles of side length 1. Determine the number of rhombuses of side length 1 bounded by line segments of the grid.
17. Show that the 7-step paths from A to B in the grid below are in 1-1 correspondence with arrangements of the letters RRRUUU.
18. How many 4-letter words can we form such that each word has 4 different letters in increasing alphabetical order (such as DIPU)?
19. How many 3-letter "words" have their middle letter later in the alphabet than both of the other two letters? (For instance, CYJ and FKF are examples of such words, but RRB and WWL are not.)
20. Compute the number of distinct paths not passing through point $(2, 2, 2)$ that travel from point $(0, 0, 0)$ to point $(4, 4, 4)$ in 12 steps, changing a coordinate by 1 at each step.
21. What is the maximum number of intersection points of diagonals inside a convex n -gon? For example, the picture at right shows a pentagon with 5 such intersection points circled.
22. Let $n = 2^{12}3^85^6$. How many divisors of n^2 are less than n ?

23. Let k be an odd number. Show that there are fewer odd divisors of $4k$ than even divisors of $4k$.
24. Annie the Ant starts at the lattice point $(0,0)$ and each minute moves 1 space up or 1 space to the right (with equal probability). Benny the Beetle starts at $(5,7)$ and each minute moves 1 space down or 1 space to the left (with equal probability). What is the probability that they meet?
25. Suppose n points are selected on the circumference of a circle and all $\binom{n}{2}$ chords connecting a pair of these points are drawn. Given that no three chords pass through the same point inside the circle, find the number of triangles that are formed by portions of the chords inside the circle and do not have any of the n points as vertices.
26. A class of 10 students took a math test. Each problem was solved by exactly 7 of the students. If the first nine students each solved 4 problems, how many problems did the tenth student solve?
27. How many nonempty subsets of $\{1,2,3,\dots,12\}$ have the property that the sum of the largest element and the smallest element is 13?
28. What is the probability that a randomly chosen divisor of 30^{39} is a multiple of 30^{29} ?
29. Consider the sequence 1, 4, 5, 16, 17, 20, 21, 64, 65,..., which is formed by including only positive integers that can be expressed as the sum of distinct powers of 4. What is the 50^{th} term in this sequence?
30. On the Cartesian plane, Johnny wants to travel from $(0,0)$ to $(5, 1)$, and he wants to pass through all twelve lattice points (x, y) such that $0 \leq x \leq 5$ and $0 \leq y \leq 1$. On each step, Johnny can go from one point to any other point via the straight line segment connecting the two points. How many paths are there from $(0,0)$ to $(5, 1)$, passing through all 12 points, such that the path never crosses itself? One such path is shown below.
31. Eight knights are randomly placed on a chessboard (not necessarily on distinct squares). A knight on a given square attacks all the squares that can be reached by moving either (1) two squares up or down followed by one square left or right, or (2) two squares left or right followed by one square up or down. Find the probability that every square, occupied or not, is attacked by some knight.
32. How many 4×4 matrices whose entries are each 1 or -1 are such that the sum of the entries in each row is 0 and the sum of the entries in each column is 0?
33. Consider the set $S = \{1,2,3,4,5,\dots,100\}$. How many subsets of this set with 2 or more elements satisfy: the terms of the subset form an arithmetic sequence, and we cannot include another element from S with this subset to form an even longer arithmetic sequence with the same common difference as the original sequence?

34. In a classroom, 34 students are seated in 5 rows of 7 chairs. The place at the center of the room is unoccupied. A teacher decides to reassign the seats such that each student will occupy a chair adjacent to his/her present one (ie. move one desk forward, back, left, or right). In how many ways can this reassignment be made?
35. Is there a 2000-element subset of the set $\{1, 2, 3, \dots, 3000\}$ such that no element in the subset is exactly double another element of the subset? Hints: 113, 155
36. Three numbers, a_1, a_2, a_3 are drawn randomly, without replacement, from the set $\{1, 2, 3, \dots, 1000\}$. Three other numbers, b_1, b_2, b_3 are then drawn randomly, without replacement, from the remaining set of 997 numbers. What is the probability that, after suitable rotation, a brick of dimensions $a_1 \times a_2 \times a_3$ can be enclosed in a box of dimension $b_1 \times b_2 \times b_3$, with the sides of the brick parallel to the sides of the box?